

Experiment 17: Mobile Price Prediction

Aim

To implement a **Mobile Price Prediction model** using an appropriate machine learning algorithm.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the mobile price dataset.

Step 3: Check and handle missing values.

Step 4: Separate the input features and target variable.

Step 5: Split the dataset into training and testing sets.

Step 6: Create a suitable classification model.

Step 7: Train the model using the training data.

Step 8: Predict the mobile price category for the test data.

Step 9: Calculate and display the accuracy.

Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = pd.read_csv("mobile_price.csv")

X = data.drop("Price", axis=1)
y = data["Price"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Prices:", y_pred)
print("R2 Score:", r2_score(y_test, y_pred))
```

Output

```
... Predicted Prices: [ 380.96371652 1068.71469198 269.85915395]  
R2 Score: 0.9744668670972908
```

Result

Thus, the **Mobile Price Prediction model** was successfully implemented using Linear Regression, and the prices of mobile phones were predicted based on their specifications.